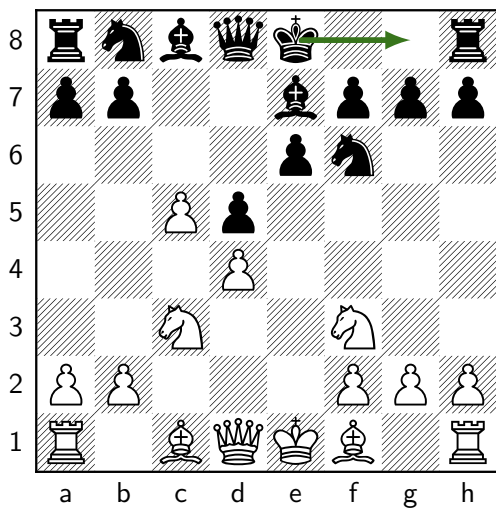


Melanie-BASE — Caro-Kann, Panov

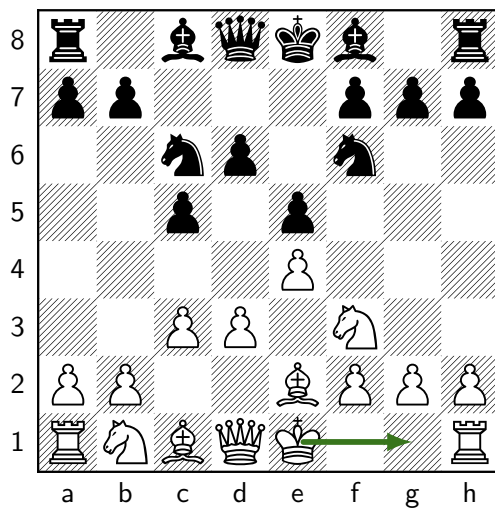
Melanie (Deep_Vision)

Abstract

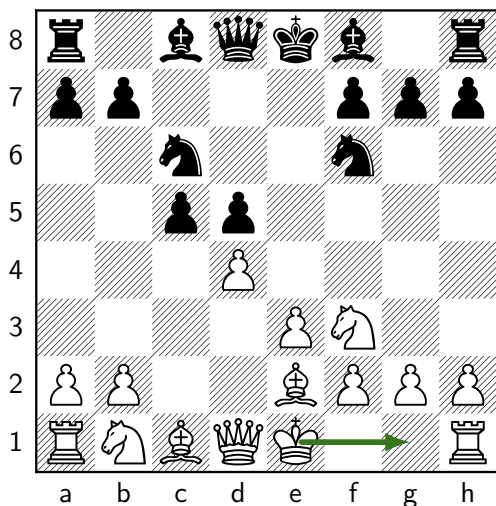
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



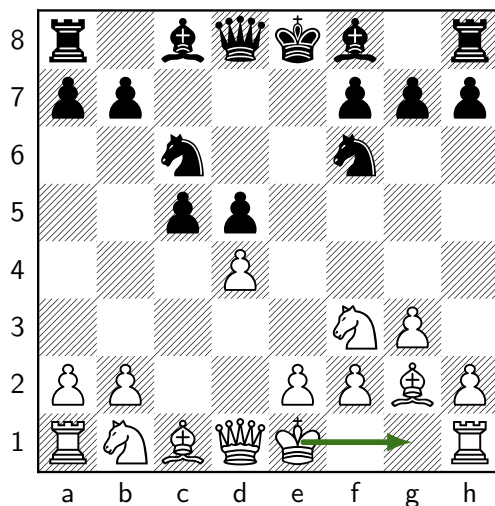
(a)



(b)

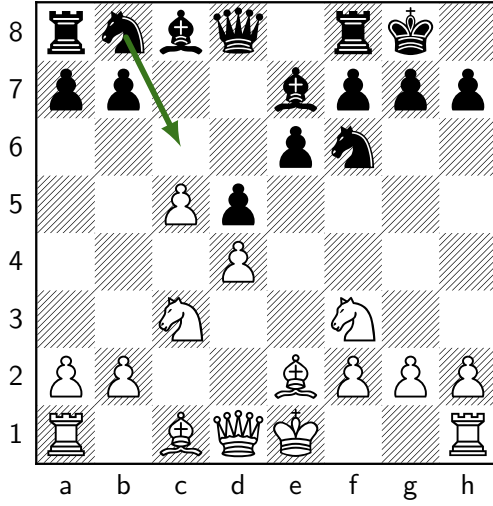


(c)

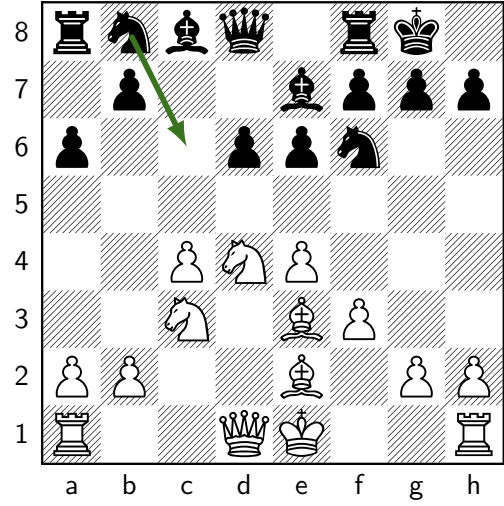


(d)

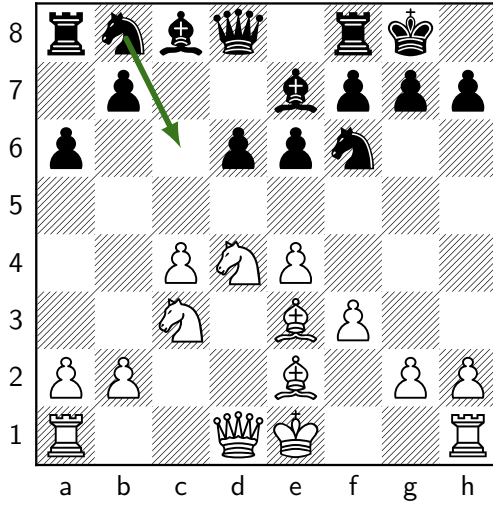
Figure 1: **Move 7, Black: 0-0.** (a) Our position; Melanie plays 0-0 (e8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1396` (ply 10) [structural distance $d=28$, QAP obj=41]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Pa7* as their *Nb1*, our *nf3* as their *Bc1*; transferred move e8g8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2417` (ply 12) [structural distance $d=14$, QAP obj=36]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Nf6* as their *Rh1*, our *Rh8* as their *Nf3*; transferred move e8g8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1124` (ply 12) [structural distance $d=33$, QAP obj=30]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Bc8* as their *Nb1*, our *bc1* as their *Pg3*; transferred move e8g8 (**endorses this move**). Synthesis: 18 retrieved analogs transferred to 0-0 (summed QAP objective 530, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



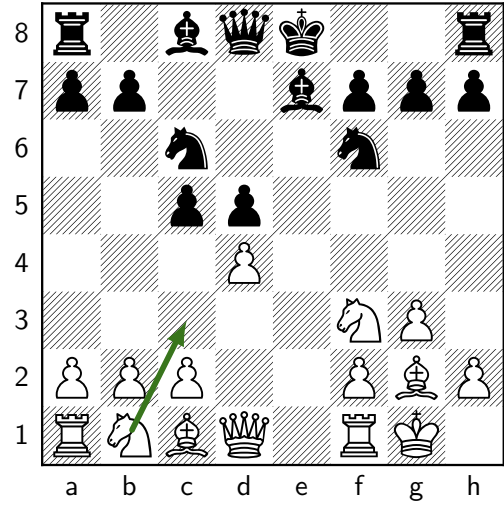
(a)



(b)

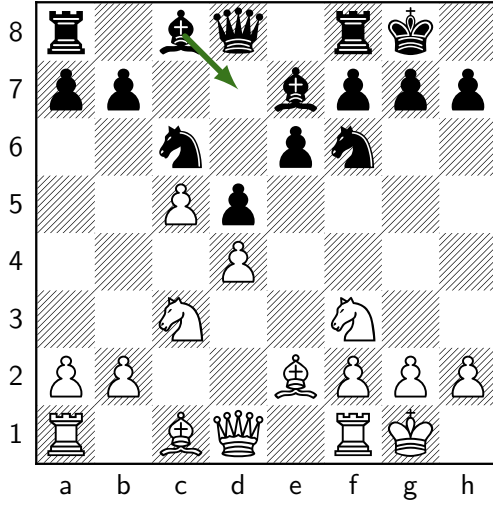


(c)

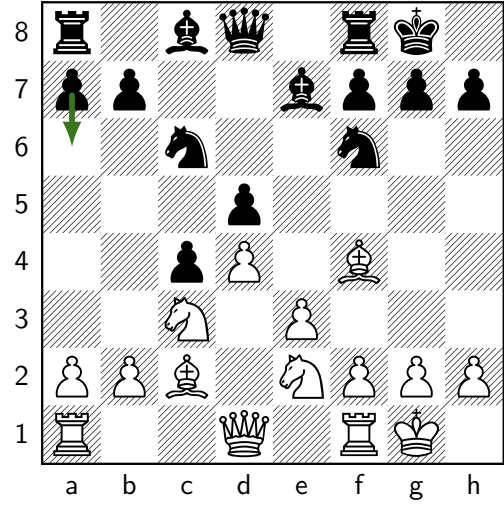


(d)

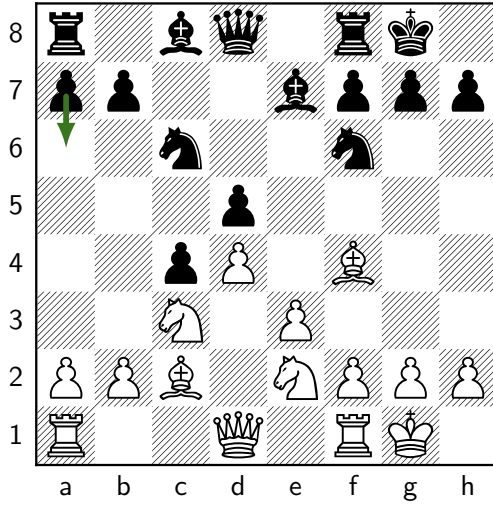
Figure 2: **Move 8, Black: Nc6.** (a) Our position; Melanie plays Nc6 (b8c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2684` (ply 17) [structural distance $d=33$, QAP obj=45]: the master played Nc6; under μ our Nb1 plays the role of that game’s Nb1; cross-type roles: our Qd8 as their Bc8, our ph2 as their Qd8; transferred move b8c6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#2685` (ply 17) [structural distance $d=33$, QAP obj=45]: the master played Nc6; under μ our Nb1 plays the role of that game’s Nb1; cross-type roles: our Qd8 as their Bc8, our ph2 as their Qd8; transferred move b8c6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#877` (ply 14) [structural distance $d=36$, QAP obj=37]: the master played Nc3; under μ our Nb1 plays the role of that game’s Nb1; cross-type roles: our Bc8 as their Pa2, our nc3 as their pc5; transferred move b8c6 (**endorses this move**). Synthesis: 17 retrieved analogs transferred to Nc6 (summed QAP objective 580, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



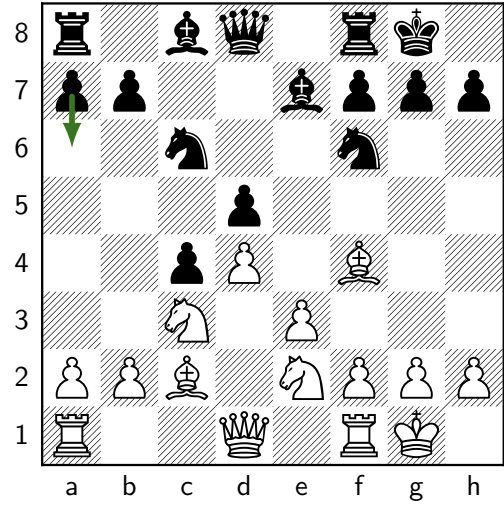
(a)



(b)

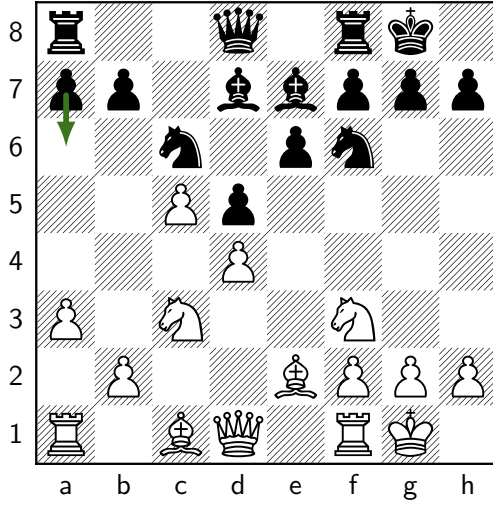


(c)

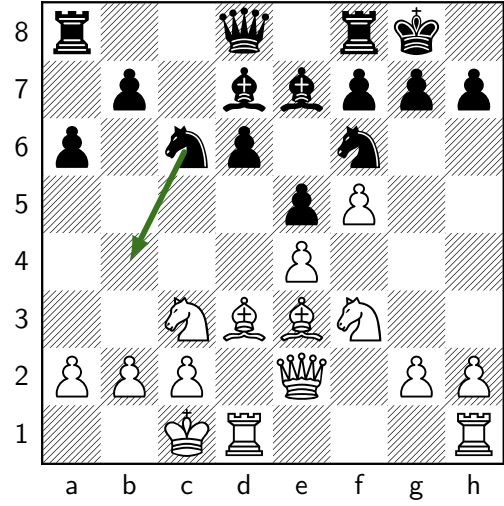


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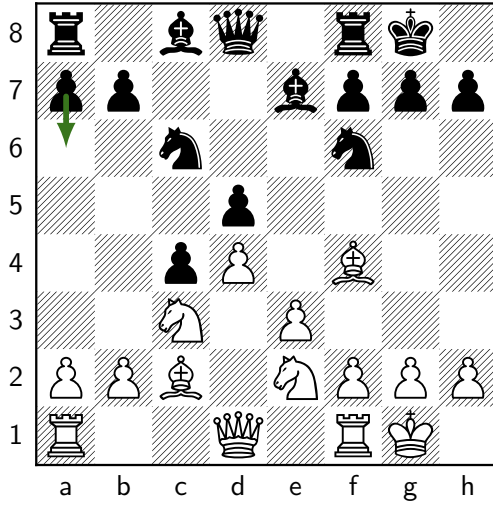
Figure 3: **Move 9, Black: Bd7**. (a) Our position; Melanie plays Bd7 (c8d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1586` (ply 19) [structural distance $d=43$, QAP obj=56]: the master played a6; under μ our *Bc1* plays the role of that game’s *Pa2* (a **cross-type** correspondence); cross-type roles: our *be2* as their *Ra8*, our *Nf6* as their *Be7*; transferred move c8d7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#2787` (ply 19) [structural distance $d=43$, QAP obj=56]: the master played a6; under μ our *Bc1* plays the role of that game’s *Pa2* (a **cross-type** correspondence); cross-type roles: our *be2* as their *Ra8*, our *Nf6* as their *Be7*; transferred move c8d7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#2874` (ply 19) [structural distance $d=43$, QAP obj=56]: the master played a6; under μ our *Bc1* plays the role of that game’s *Pa2* (a **cross-type** correspondence); cross-type roles: our *be2* as their *Ra8*, our *Nf6* as their *Be7*; transferred move c8d7 (**endorses this move**). Synthesis: 13 retrieved analogs transferred to Bd7 (summed QAP objective 700, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



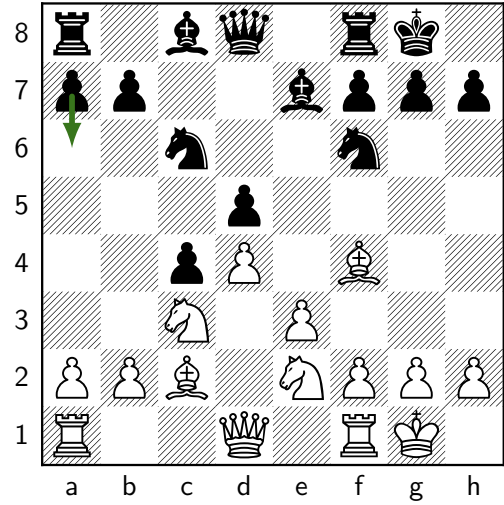
(a)



(b)

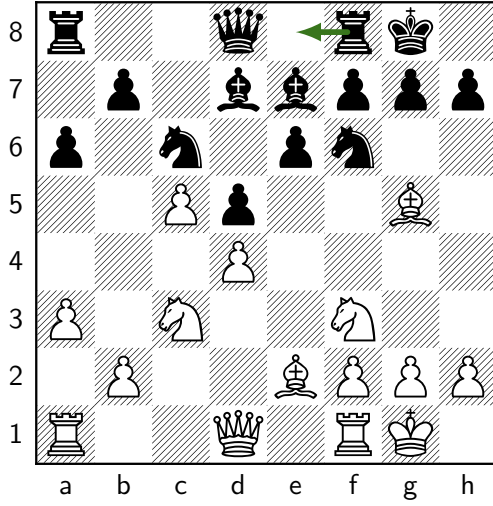


(c)

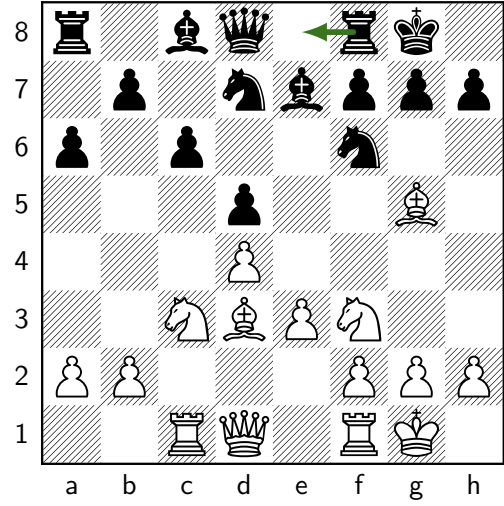


(d)

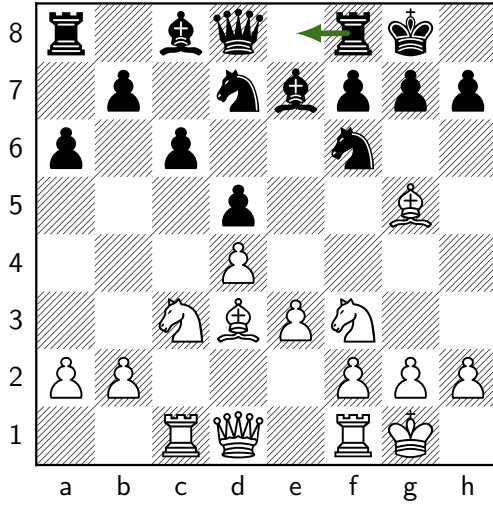
Figure 4: **Move 10, Black: a6.** (a) Our position; Melanie plays a6 (a7a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3407` (ply 23) [structural distance $d=42$, QAP obj=66]: the master played Nb4; under μ our *Pa2* plays the role of that game’s *Nc3* (a **cross-type** correspondence); cross-type roles: our *Nf6* as their *Rf8*, our *Nc6* as their *Bd7*; transferred move a7a6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1586` (ply 19) [structural distance $d=49$, QAP obj=53]: the master played a6; under μ our *Pa2* plays the role of that game’s *Pa2*; cross-type roles: our *bc1* as their *Ra8*, our *ra1* as their *Bc8*; transferred move a7a6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2787` (ply 19) [structural distance $d=49$, QAP obj=53]: the master played a6; under μ our *Pa2* plays the role of that game’s *Pa2*; cross-type roles: our *bc1* as their *Ra8*, our *ra1* as their *Bc8*; transferred move a7a6 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to a6 (summed QAP objective 443, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



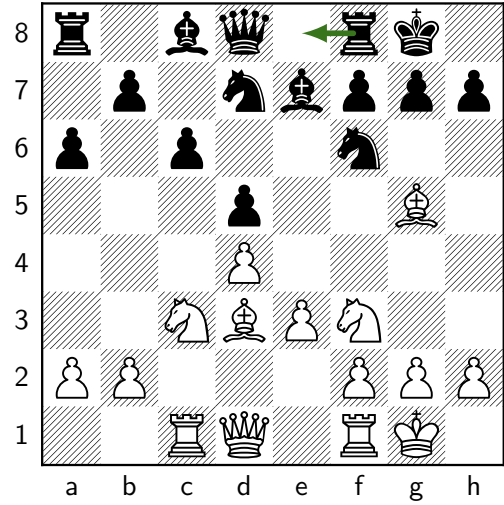
(a)



(b)

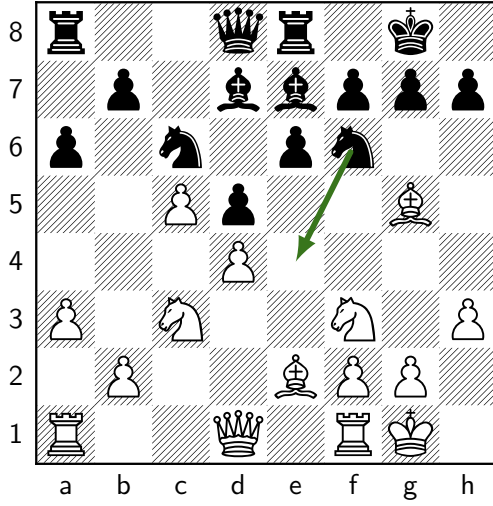


(c)

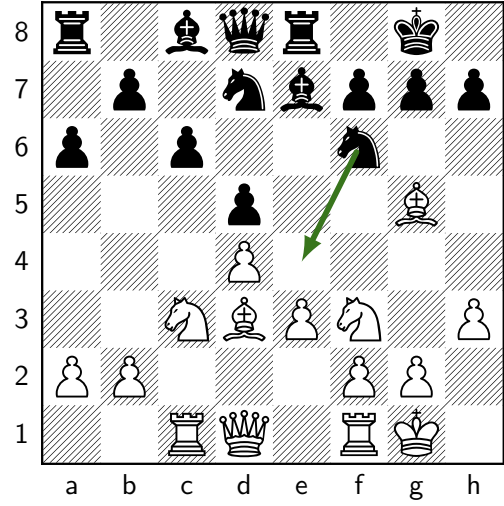


(d)

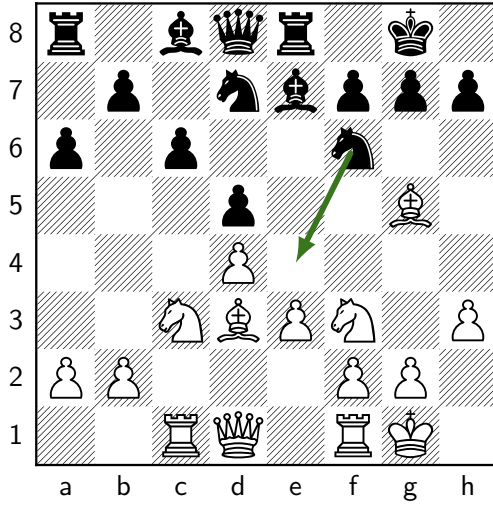
Figure 5: **Move 11, Black: Re8.** (a) Our position; Melanie plays **Re8** (**f8e8**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1170` (ply 19) [structural distance $d=52$, QAP obj=32]: the master played **Re8**; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *Qd8* as their *Ra8*, our *nc3* as their *Qd8*; transferred move **f8e8** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1171` (ply 19) [structural distance $d=52$, QAP obj=32]: the master played **Re8**; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *Qd8* as their *Ra8*, our *nc3* as their *Qd8*; transferred move **f8e8** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1517` (ply 19) [structural distance $d=52$, QAP obj=32]: the master played **Re8**; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *Qd8* as their *Ra8*, our *nc3* as their *Qd8*; transferred move **f8e8** (**endorses this move**). Synthesis: 9 retrieved analogs transferred to **Re8** (summed QAP objective 289, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



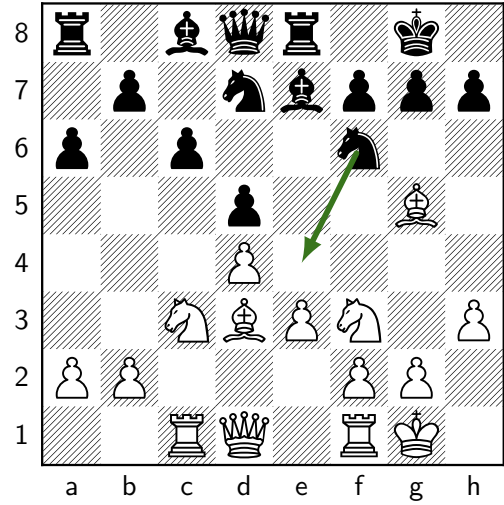
(a)



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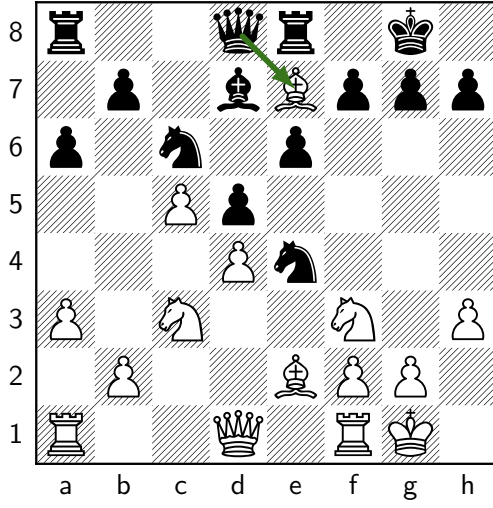


(c)

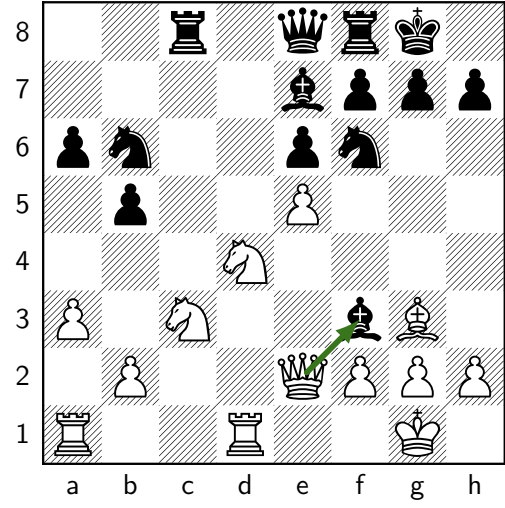


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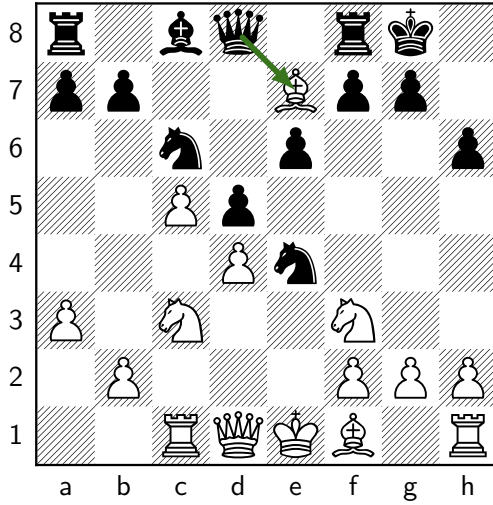
Figure 6: **Move 12, Black: Ne4.** (a) Our position; Melanie plays Ne4 (f6e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1517` (ply 21) [structural distance $d=65$, QAP obj=34]: the master played Ne4; under μ our Nf3 plays the role of that game’s Nf3; cross-type roles: our Pa6 as their Bc8, our nc3 as their Qd8; transferred move f6e4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3613` (ply 21) [structural distance $d=65$, QAP obj=34]: the master played Ne4; under μ our Nf3 plays the role of that game’s Nf3; cross-type roles: our Pa6 as their Bc8, our nc3 as their Qd8; transferred move f6e4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2985` (ply 21) [structural distance $d=65$, QAP obj=34]: the master played Ne4; under μ our Nf3 plays the role of that game’s Nf3; cross-type roles: our Pa6 as their Bc8, our nc3 as their Qd8; transferred move f6e4 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Ne4 (summed QAP objective 155, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



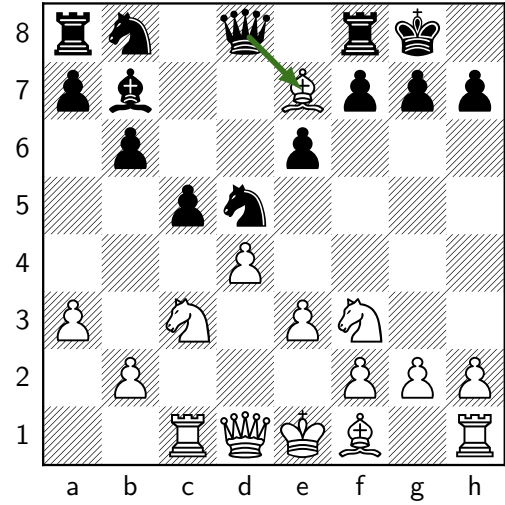
(a)



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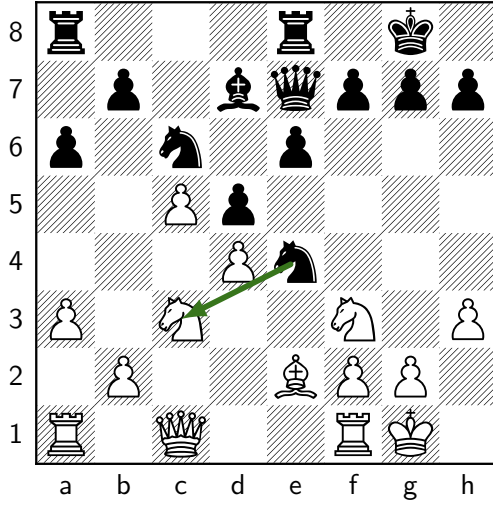


(c)

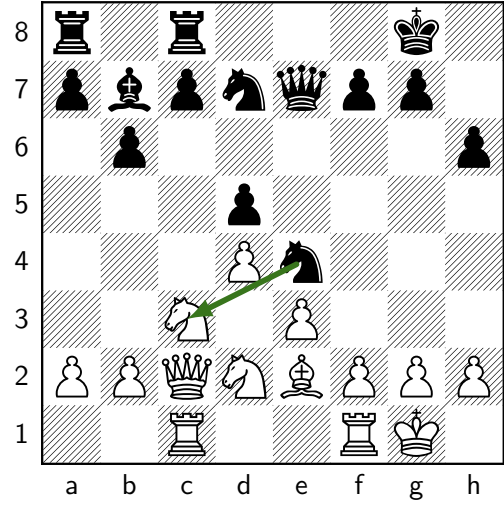


(d)

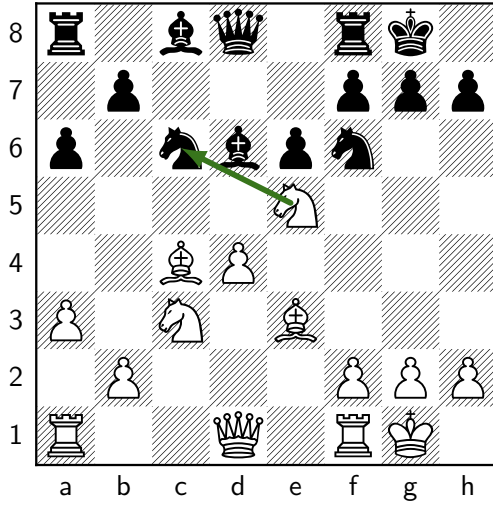
Figure 7: **Move 13, Black: Qxe7.** (a) Our position; Melanie plays Qxe7 (d8e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#2141` (ply 36) [structural distance $d=108$, QAP obj=71]: the master played Qxf3; under μ our Qd1 plays the role of that game's Qe2; cross-type roles: our Pg7 as their Rd1, our Re8 as their Pg2; transferred move d8e7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#63` (ply 23) [structural distance $d=80$, QAP obj=64]: the master played Qxe7; under μ our Qd1 plays the role of that game's Qd1; cross-type roles: our Pg7 as their Ra8, our pc5 as their Bc8; transferred move d8e7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#2735` (ply 19) [structural distance $d=108$, QAP obj=58]: the master played Qxe7; under μ our Qd1 plays the role of that game's Qd1; cross-type roles: our Bd7 as their Nb8, our Qd8 as their Rf8; transferred move d8e7 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Qxe7 (summed QAP objective 376, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



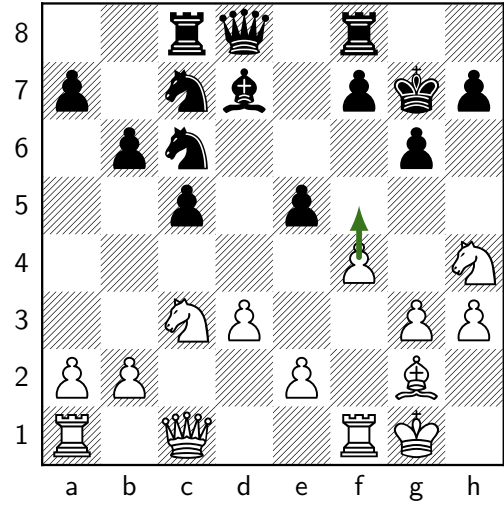
(a)



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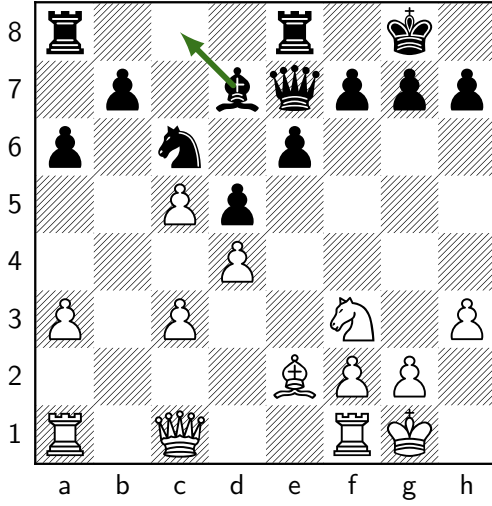


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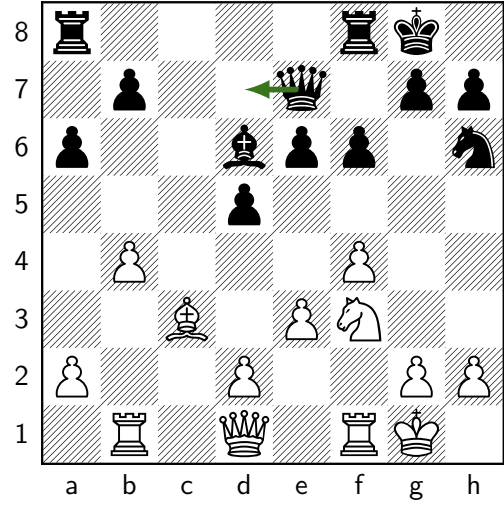


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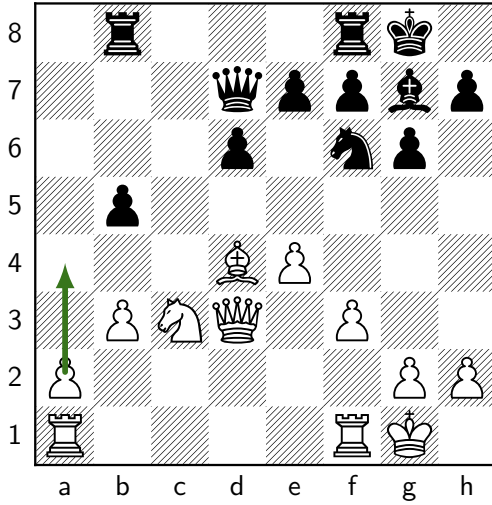
Figure 8: **Move 14, Black: Nxc3.** (a) Our position; Melanie plays Nxc3 (e4c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#1223 (ply 27) [structural distance $d=68$, QAP obj=137]: the master played Nxc3; under μ our Ne5 plays the role of that game’s Ne5; cross-type roles: our Pe6 as their Bb7, our Bd7 as their Pc7; transferred move e4c3 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#318 (ply 22) [structural distance $d=79$, QAP obj=66]: the master played Nxc6; under μ our Ne5 plays the role of that game’s Ne5; cross-type roles: our qc1 as their Be3, our nc3 as their bd6; transferred move e4c3 (**endorses this move**). (d) Retrieved from ccr1_top_p0003#652 (ply 30) [structural distance $d=78$, QAP obj=47]: the master played f5; under μ our Ne5 plays the role of that game’s Pf4 (a **cross-type** correspondence); cross-type roles: our Pb7 as their Ra1, our rf1 as their Qc1; transferred move e4c3 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Nxc3 (summed QAP objective 277, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



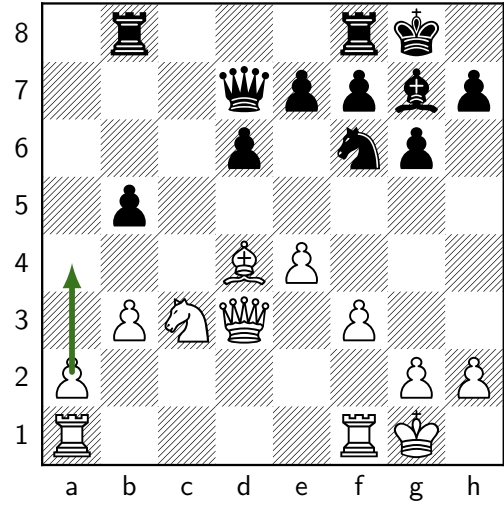
(a)



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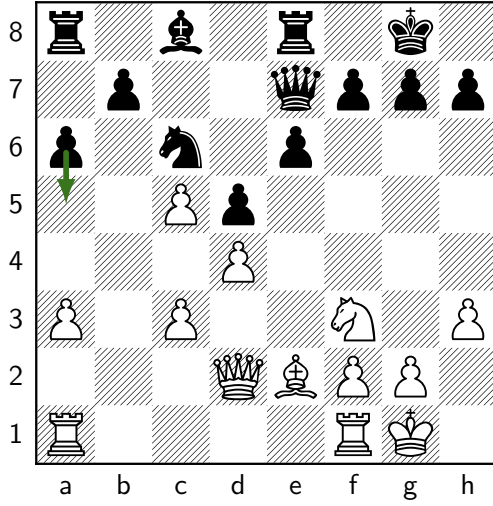


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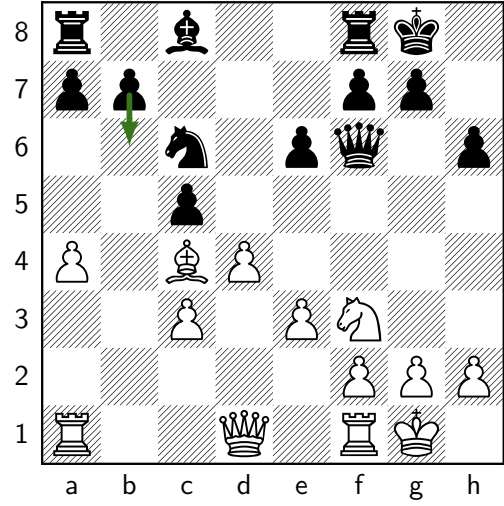


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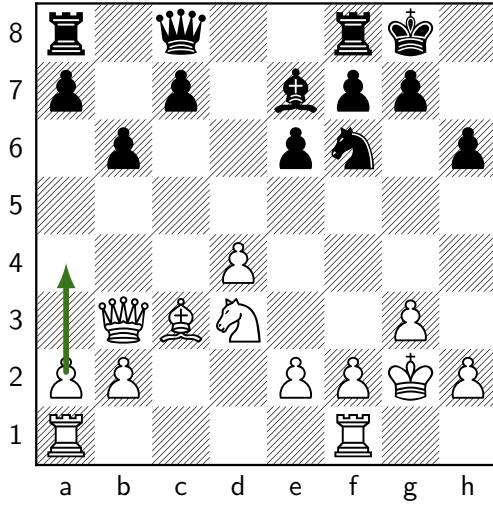
Figure 9: **Move 15, Black: Bc8.** (a) Our position; Melanie plays Bc8 (d7c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2665` (ply 33) [structural distance $d=51$, QAP obj=51]: the master played Qd7; under μ our Bd2 plays the role of that game’s Qe2 (a **cross-type** correspondence); cross-type roles: our Qe7 as their Bd6, our ra1 as their Pe6; transferred move d7c8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2390` (ply 30) [structural distance $d=51$, QAP obj=26]: the master played a4; under μ our Bd2 plays the role of that game’s Pa2 (a **cross-type** correspondence); cross-type roles: our Qe7 as their Nc3, our Nc6 as their Bd4; transferred move d7c8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2848` (ply 30) [structural distance $d=51$, QAP obj=26]: the master played a4; under μ our Bd2 plays the role of that game’s Pa2 (a **cross-type** correspondence); cross-type roles: our Qe7 as their Nc3, our Nc6 as their Bd4; transferred move d7c8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Bc8 (summed QAP objective 114, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



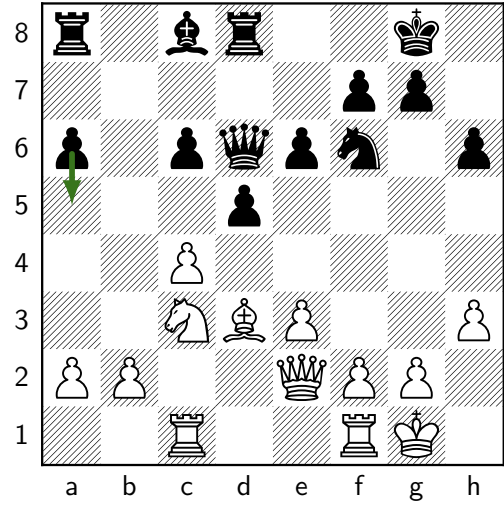
(a)



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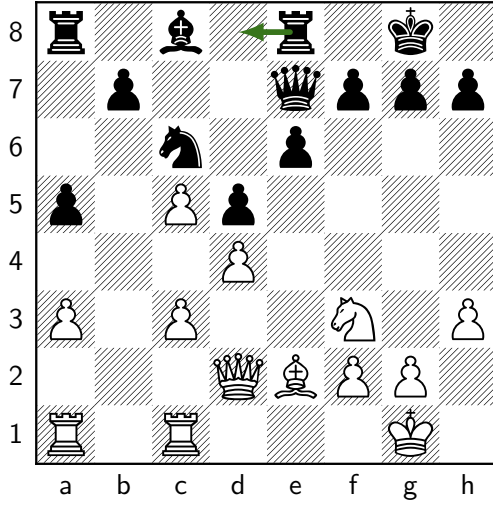


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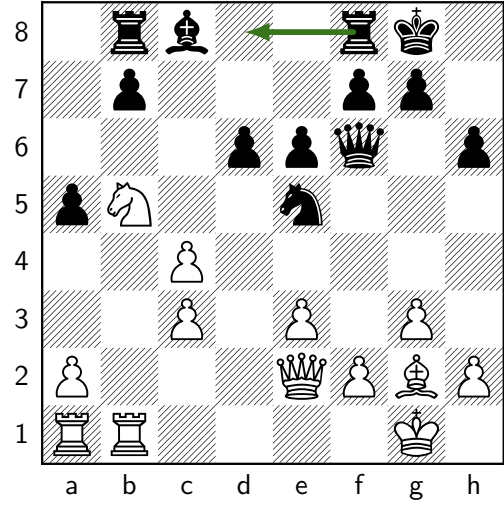


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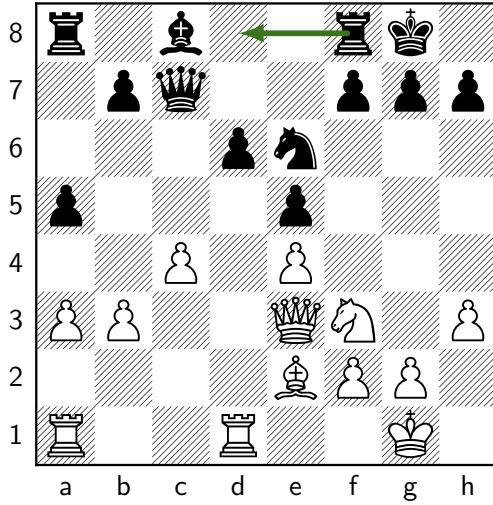
Figure 10: **Move 16, Black: a5**. (a) Our position; Melanie plays a5 (a6a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#53` (ply 23) [structural distance $d=57$, QAP obj=41]: the master played b6; under μ our $Pa3$ plays the role of that game’s $Pb2$; cross-type roles: our $Nc6$ as their $Qf6$, our $Nc6$ as their $Pc5$; transferred move a6a5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3257` (ply 30) [structural distance $d=60$, QAP obj=31]: the master played a4; under μ our $Pa3$ plays the role of that game’s $Pa2$; cross-type roles: our $Pg7$ as their $Ra1$, our $Qe7$ as their $Rf1$; transferred move a6a5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#768` (ply 29) [structural distance $d=53$, QAP obj=24]: the master played a5; under μ our $Pa3$ plays the role of that game’s $Pa3$; cross-type roles: our $Qe7$ as their $Nf6$, our $Ra8$ as their $Ph6$; transferred move a6a5 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to a5 (summed QAP objective 139, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



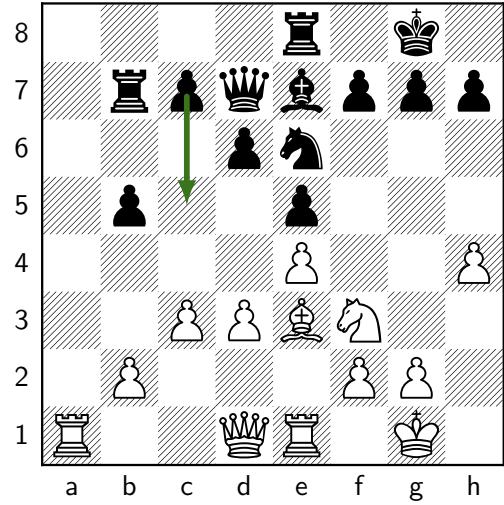
(a)



(b)

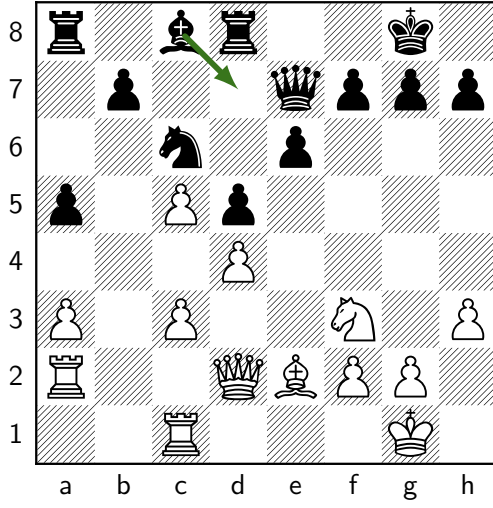


(c)

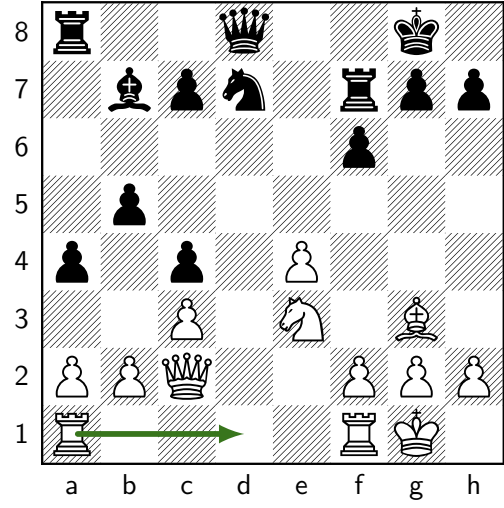


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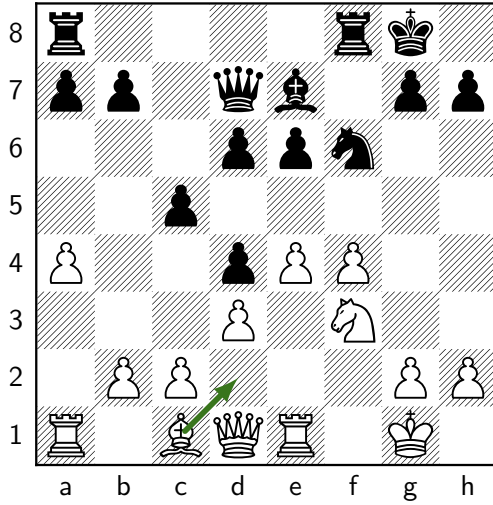
Figure 11: **Move 17, Black: Rd8.** (a) Our position; Melanie plays Rd8 (e8d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#560` (ply 35) [structural distance $d=61$, QAP obj=42]: the master played Rd8; under μ our *Re1* plays the role of that game’s *Rf1*; cross-type roles: our *Pb7* as their *Rb8*, our *rc1* as their *Bc8*; transferred move e8d8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3430` (ply 33) [structural distance $d=65$, QAP obj=36]: the master played Rd8; under μ our *Re1* plays the role of that game’s *Rf1*; cross-type roles: our *Pa5* as their *Bc8*, our *be2* as their *ph3*; transferred move e8d8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3015` (ply 35) [structural distance $d=65$, QAP obj=34]: the master played c5; under μ our *Re1* plays the role of that game’s *Pc2* (a **cross-type** correspondence); cross-type roles: our *Bc8* as their *Re8*, our *pc3* as their *Rb7*; transferred move e8d8 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Rd8 (summed QAP objective 138, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



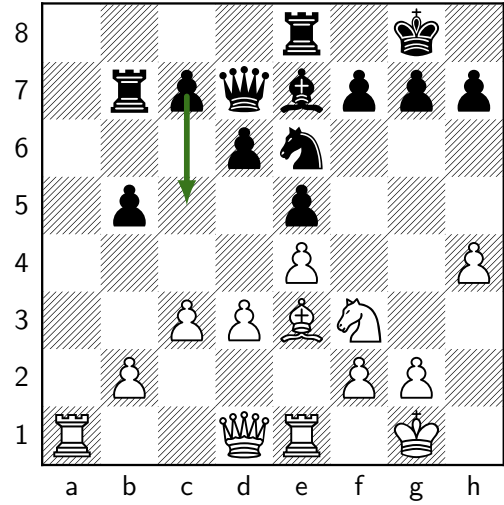
(a)



(b)

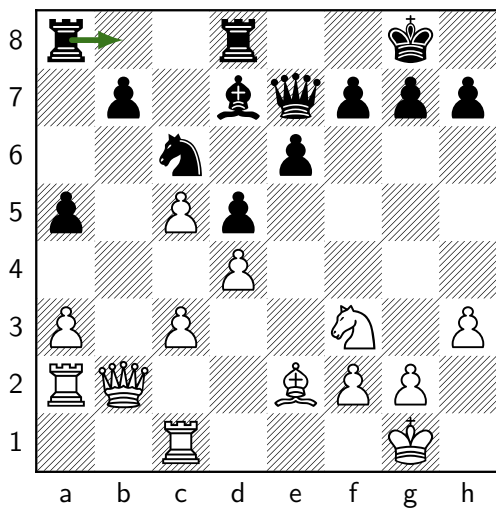


(c)

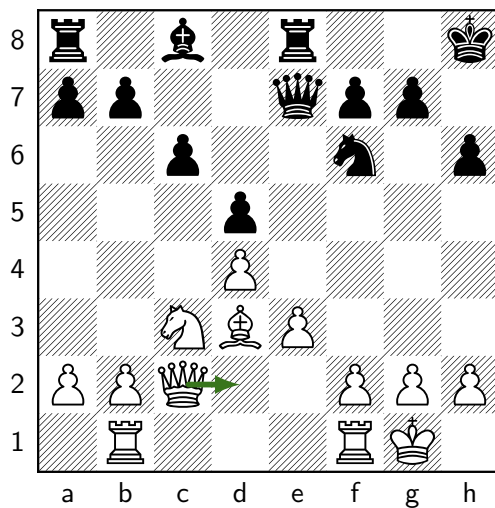


(d)

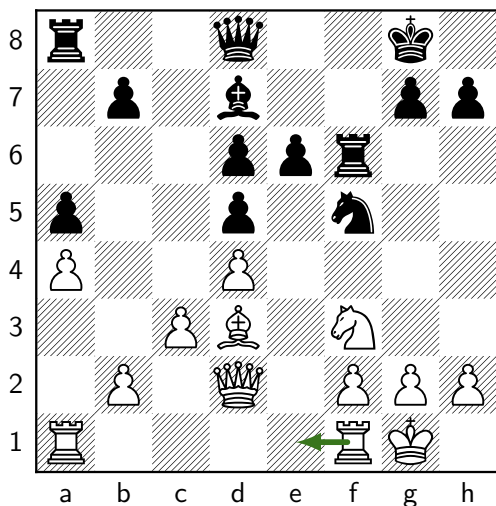
Figure 12: **Move 18, Black: Bd7.** (a) Our position; Melanie plays Bd7 (c8d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0003#717` (ply 38) [structural distance $d=59$, QAP obj=43]: the master played Rad1; under μ our Bc1 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our rc1 as their Pc3, our qd2 as their Bg3; transferred move c8d7 (**endorses this move**). (c) Retrieved from `ccl1_top_p0000#3938` (ply 26) [structural distance $d=69$, QAP obj=39]: the master played Bd2; under μ our Bc1 plays the role of that game’s Bc1; cross-type roles: our Pe6 as their Qd1, our pc3 as their Re1; transferred move c8d7 (**endorses this move**). (d) Retrieved from `ccl1_top_p0002#3015` (ply 35) [structural distance $d=65$, QAP obj=36]: the master played c5; under μ our Bc1 plays the role of that game’s Pc2 (a **cross-type** correspondence); cross-type roles: our Pa5 as their Qd7, our Qe7 as their Be7; transferred move c8d7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Bd7 (summed QAP objective 139, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



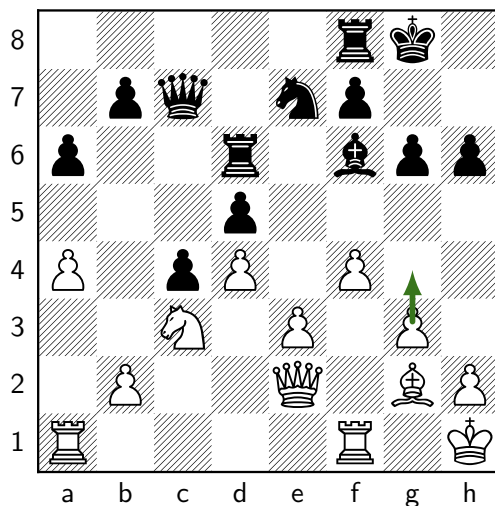
(a)



(b)

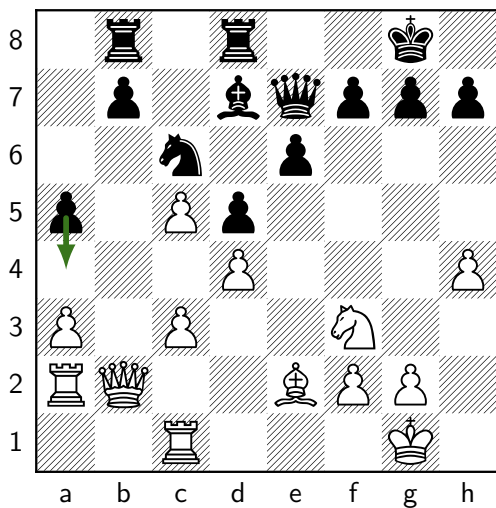


(c)

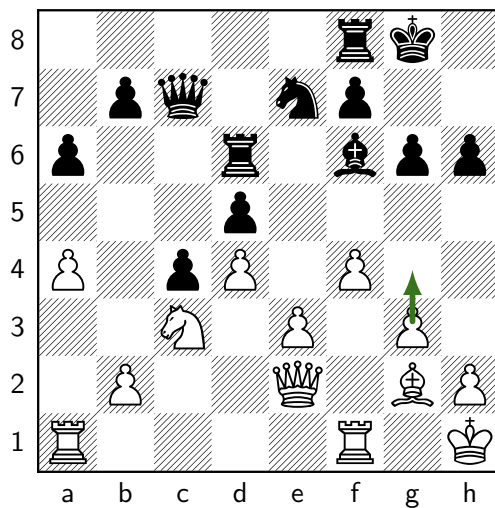


(d)

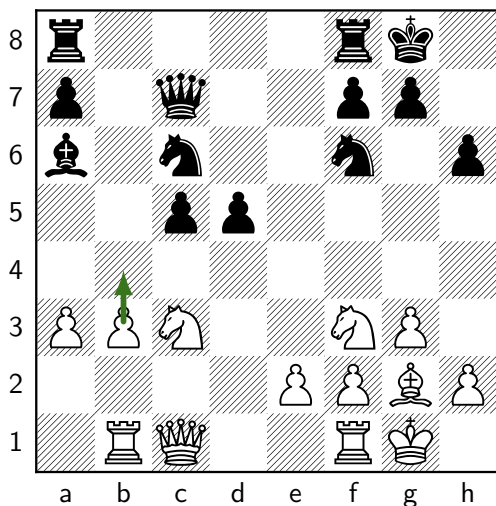
Figure 13: **Move 19, Black: Rab8.** (a) Our position; Melanie plays Rab8 (a8b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#702` (ply 32) [structural distance $d=51$, QAP obj=35]: the master played Qd2; under μ our Ra1 plays the role of that game’s Qc2 (a **cross-type** correspondence); cross-type roles: our ph3 as their Rf1, our nf3 as their Pf2; transferred move a8b8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1294` (ply 36) [structural distance $d=54$, QAP obj=14]: the master played Rfe1; under μ our Ra1 plays the role of that game’s Rf1; cross-type roles: our pc3 as their Ra1, our Bd7 as their Pb2; transferred move a8b8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2663` (ply 38) [structural distance $d=47$, QAP obj=69]: the master played g4; under μ our Pb2 plays the role of that game’s Pg3; cross-type roles: our Bd7 as their Pb2, our Nc6 as their Bg2; transferred move b7b5. Synthesis: 2 retrieved analogs transferred to Rab8 (summed QAP objective 49, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



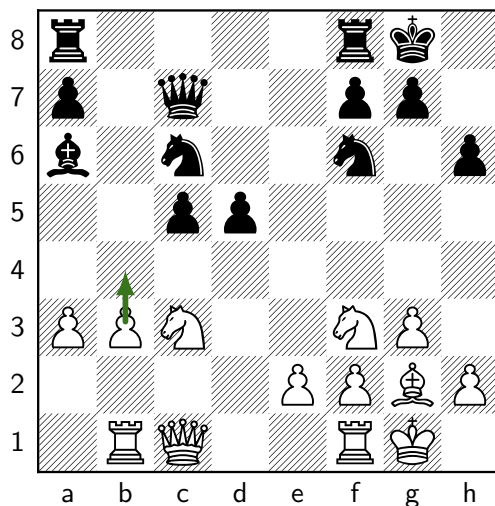
(a)



(b)

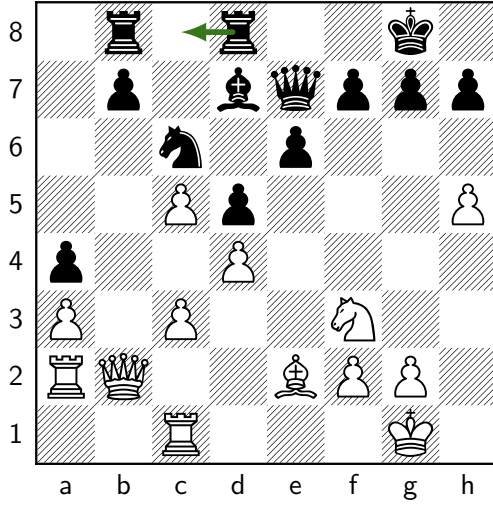


(c)

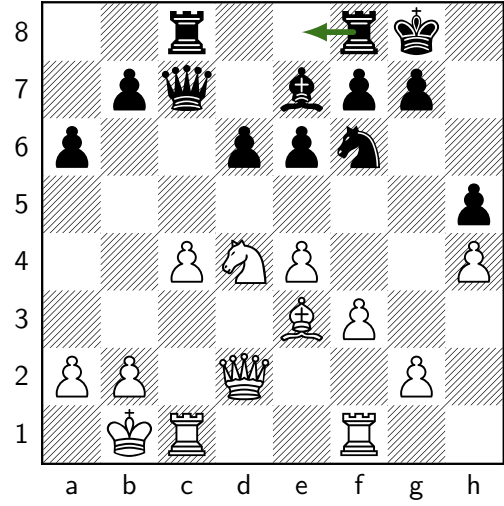


(d)

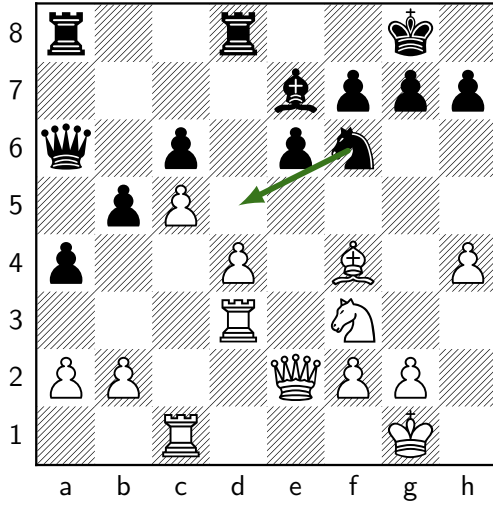
Figure 14: **Move 20, Black: a4.** (a) Our position; Melanie plays a4 (a5a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2663` (ply 38) [structural distance $d=52$, QAP obj=75]: the master played g4; under μ our Pa_4 plays the role of that game’s Pg_3 ; cross-type roles: our Bd_7 as their Pb_2 , our Nc_6 as their Bg_2 ; transferred move a5a4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2009` (ply 40) [structural distance $d=62$, QAP obj=60]: the master played b4; under μ our Pa_4 plays the role of that game’s Pb_3 ; cross-type roles: our Pf_7 as their Rb_1 , our Pe_6 as their Rf_1 ; transferred move a5a4 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#1794` (ply 40) [structural distance $d=62$, QAP obj=60]: the master played b4; under μ our Pa_4 plays the role of that game’s Pb_3 ; cross-type roles: our Pf_7 as their Rb_1 , our Pe_6 as their Rf_1 ; transferred move a5a4 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to a4 (summed QAP objective 245, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



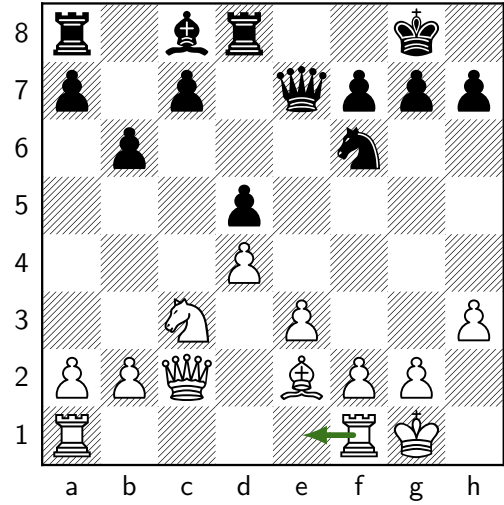
(a)



(b)

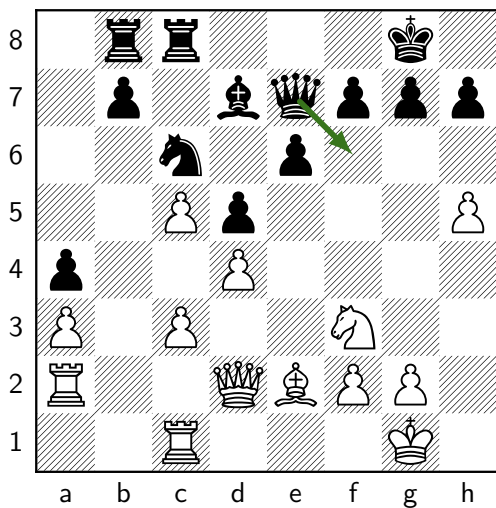


(c)

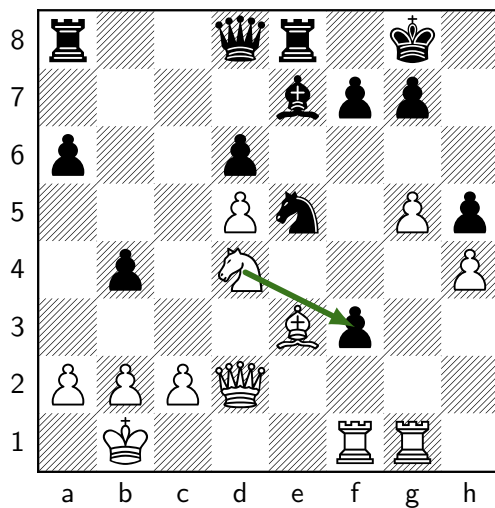


(d)

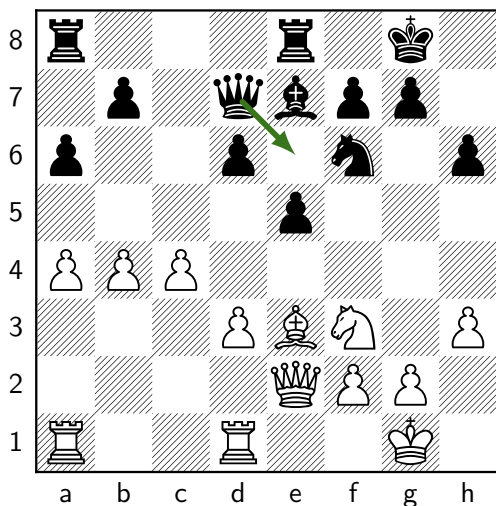
Figure 15: **Move 21, Black: Rdc8.** (a) Our position; Melanie plays Rdc8 (d8c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#2670` (ply 35) [structural distance $d=42$, QAP obj=62]: the master played Rfe8; under μ our Rd1 plays the role of that game’s Rf1; cross-type roles: our Pf7 as their Be7, our nf3 as their Pf7; transferred move d8c8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#473` (ply 41) [structural distance $d=57$, QAP obj=61]: the master played Nd5; under μ our Rd1 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Nc6 as their Rd8, our Qe7 as their Be7; transferred move d8c8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#802` (ply 30) [structural distance $d=55$, QAP obj=59]: the master played Rfe1; under μ our Rd1 plays the role of that game’s Rf1; cross-type roles: our Nc6 as their Qc2, our pg2 as their Be2; transferred move d8c8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rdc8 (summed QAP objective 234, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



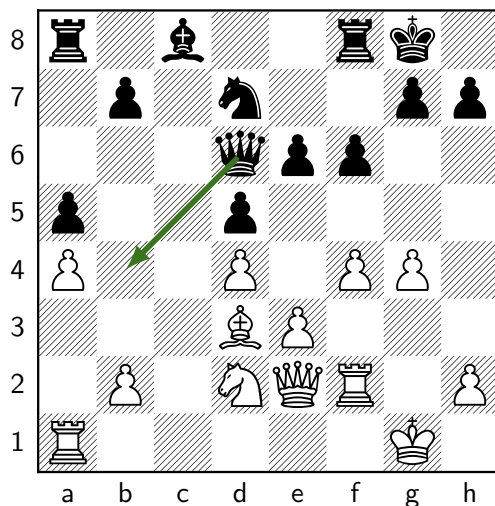
(a)



(b)

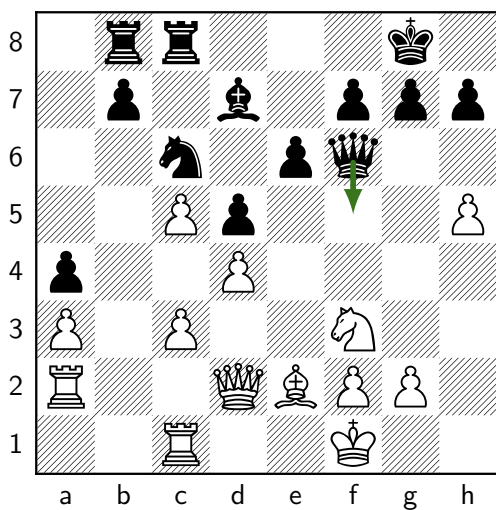


(c)

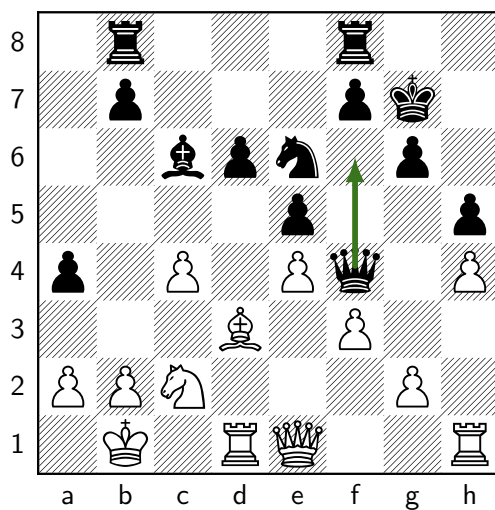


(d)

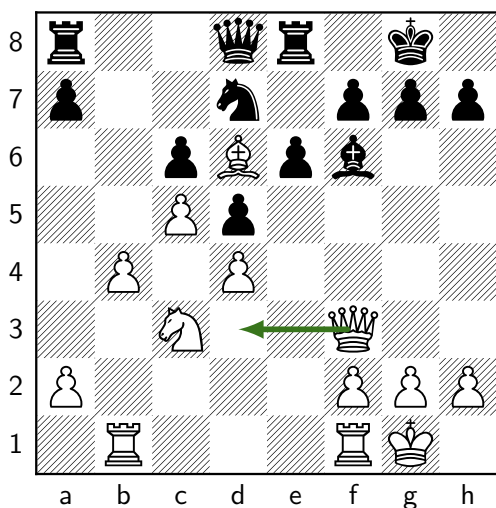
Figure 16: **Move 22, Black: Qf6.** (a) Our position; Melanie plays Qf6 (e7f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2467` (ply 44) [structural distance $d=58$, QAP obj=67]: the master played Nxf3; under μ our Qe2 plays the role of that game’s Nd4 (a **cross-type** correspondence); cross-type roles: our Qe7 as their Rf1, our Bd7 as their Rg1; transferred move e7f6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3257` (ply 35) [structural distance $d=62$, QAP obj=47]: the master played Qe6; under μ our Qe2 plays the role of that game’s Qd2; cross-type roles: our ph5 as their Ra8, our Pe6 as their Re8; transferred move e7f6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3483` (ply 35) [structural distance $d=61$, QAP obj=31]: the master played Qb4; under μ our Qe2 plays the role of that game’s Qd3; cross-type roles: our pg2 as their Ra8, our Rc8 as their Bc8; transferred move e7f6 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Qf6 (summed QAP objective 199, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



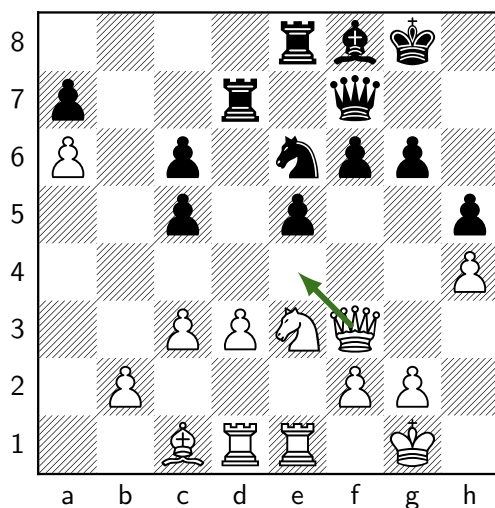
(a)



(b)

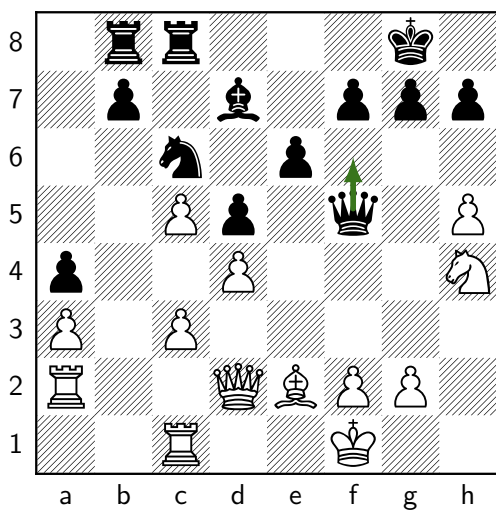


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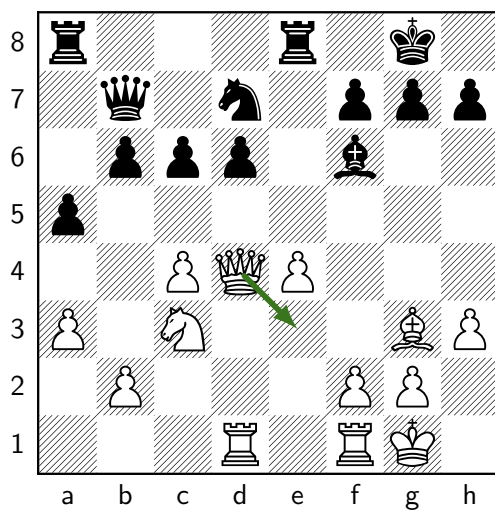


(d)

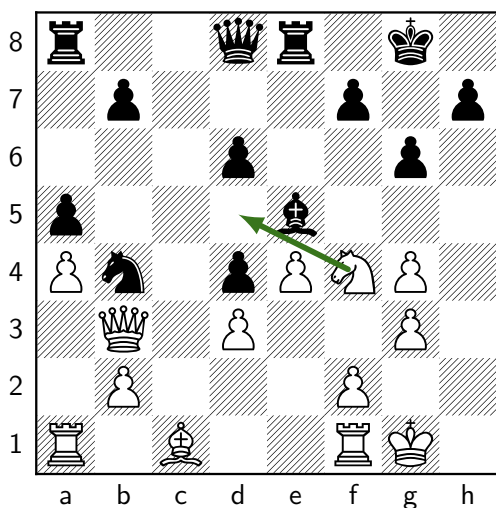
Figure 17: **Move 23, Black: Qf5.** (a) Our position; Melanie plays Qf5 (f6f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1919` (ply 43) [structural distance $d=64$, QAP obj=59]: the master played Qf6; under μ our Qf3 plays the role of that game’s Qf5; cross-type roles: our Pb7 as their Rf8, our Nc6 as their Bc6; transferred move f6f5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3200` (ply 30) [structural distance $d=66$, QAP obj=55]: the master played Qd3; under μ our Qf3 plays the role of that game’s Qf3; cross-type roles: our Bd7 as their Pa2, our pc5 as their Bd6; transferred move f6f5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2072` (ply 54) [structural distance $d=66$, QAP obj=30]: the master played Qe4; under μ our Qf3 plays the role of that game’s Qf3; cross-type roles: our Rc8 as their Bc1, our Pb7 as their Re1; transferred move f6f5 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Qf5 (summed QAP objective 243, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



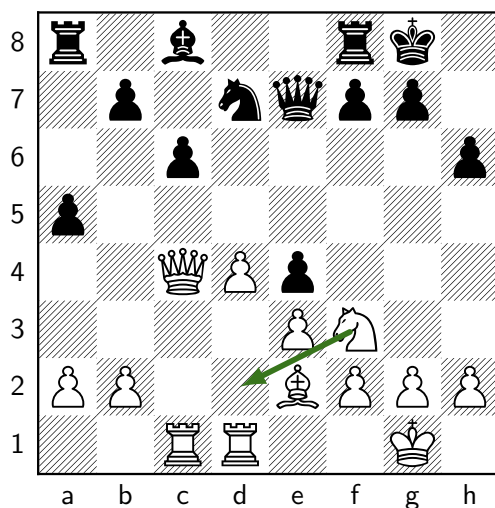
(a)



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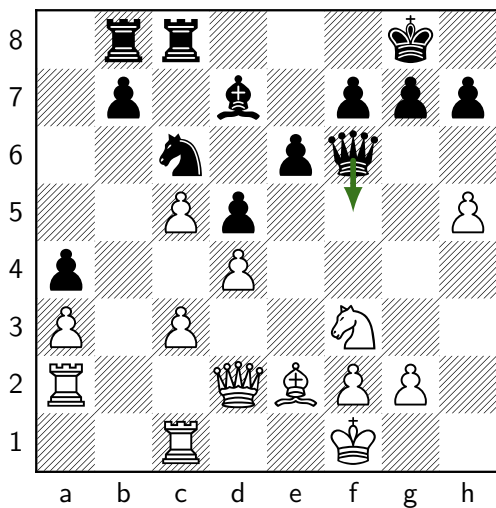


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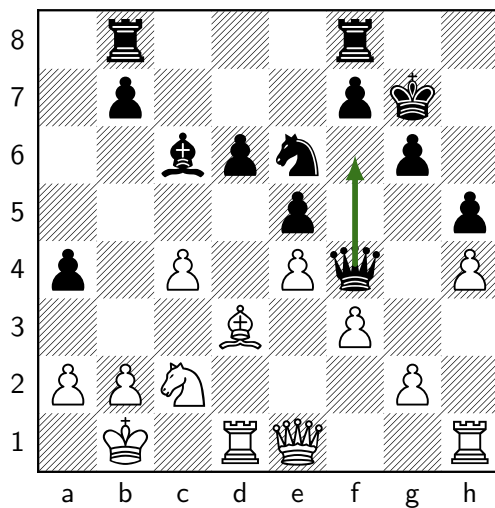


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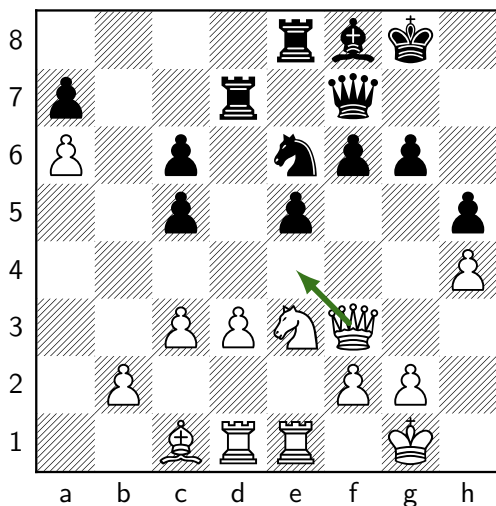
Figure 18: **Move 24, Black: Qf6**. (a) Our position; Melanie plays Qf6 (f5f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1874` (ply 38) [structural distance $d=64$, QAP obj=64]: the master played Qe3; under μ our Qf4 plays the role of that game’s Qd4; cross-type roles: our Pf7 as their Rd1, our Bd7 as their Pg2; transferred move f5f6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3569` (ply 38) [structural distance $d=71$, QAP obj=45]: the master played Nd5; under μ our Qf4 plays the role of that game’s Nf4 (a **cross-type** correspondence); cross-type roles: our pa3 as their Ra1, our ph5 as their nb4; transferred move f5f6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#877` (ply 30) [structural distance $d=74$, QAP obj=44]: the master played Nd2; under μ our Qf4 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Ph7 as their Rc1, our Pe6 as their Be2; transferred move f5f6 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Qf6 (summed QAP objective 298, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



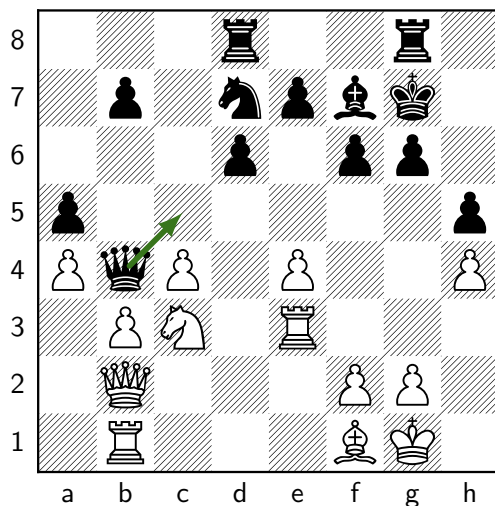
(a)



(b)

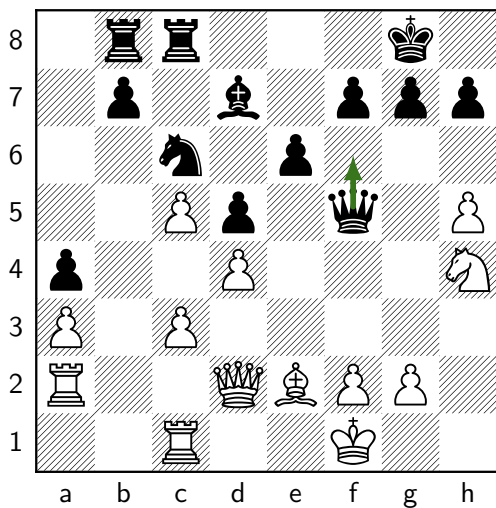


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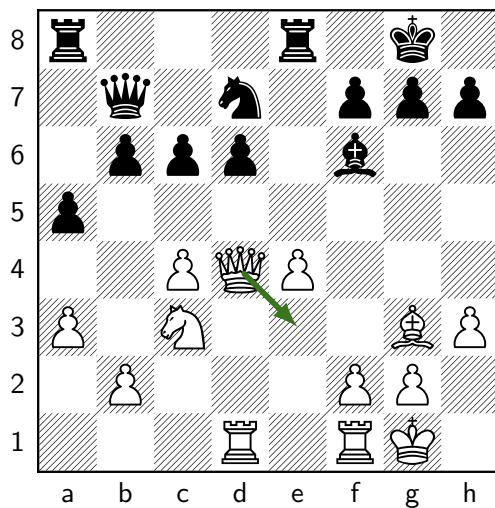


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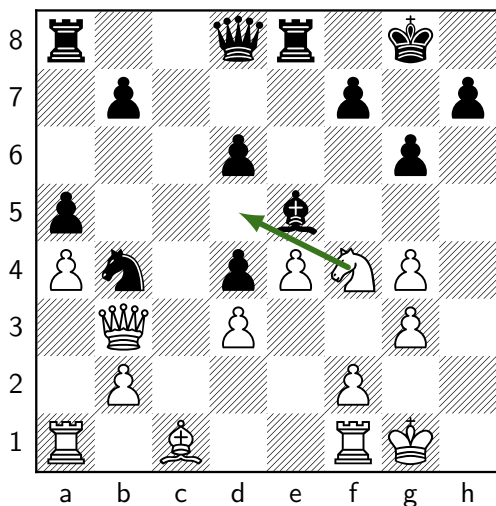
Figure 19: **Move 25, Black: Qf5.** (a) Our position; Melanie plays Qf5 (f6f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1919` (ply 43) [structural distance $d=64$, QAP obj=59]: the master played Qf6; under μ our Qf3 plays the role of that game’s Qf5; cross-type roles: our Pb7 as their Rf8, our Nc6 as their Bc6; transferred move f6f5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2072` (ply 54) [structural distance $d=66$, QAP obj=30]: the master played Qe4; under μ our Qf3 plays the role of that game’s Qf3; cross-type roles: our Rc8 as their Bc1, our Pb7 as their Re1; transferred move f6f5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1123` (ply 55) [structural distance $d=70$, QAP obj=29]: the master played Qc5; under μ our Qf3 plays the role of that game’s Qb5; cross-type roles: our Bd7 as their Rd8, our pc5 as their Nd7; transferred move f6f5 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Qf5 (summed QAP objective 188, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



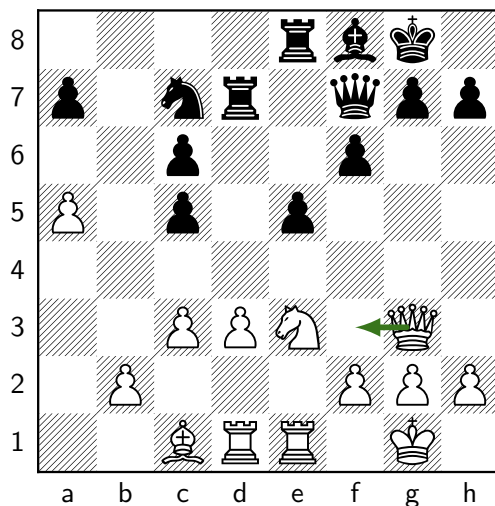
(a)



(b)



(c)



(d)

Figure 20: **Move 26, Black: Qf6**. (a) Our position; Melanie plays Qf6 (f5f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1874` (ply 38) [structural distance $d=64$, QAP obj=64]: the master played Qe3; under μ our Qf4 plays the role of that game’s Qd4; cross-type roles: our Pf7 as their Rd1, our Bd7 as their Pg2; transferred move f5f6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3569` (ply 38) [structural distance $d=71$, QAP obj=45]: the master played Nd5; under μ our Qf4 plays the role of that game’s Nf4 (a **cross-type** correspondence); cross-type roles: our pa3 as their Ra1, our ph5 as their nb4; transferred move f5f6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2072` (ply 44) [structural distance $d=64$, QAP obj=39]: the master played Qf3; under μ our Qf4 plays the role of that game’s Qg3; cross-type roles: our Rb8 as their Bc1, our nh4 as their Rd1; transferred move f5f6 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Qf6 (summed QAP objective 214, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.